

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 12. Sets

Lecture: 131.Operators On Set

Section 1: Lecture Summary

This lecture focuses on applying **operators to set objects** in Python, offering an alternative way to perform common set operations beyond using the standard set methods. The instructor revisits previously learned set operations—union, intersection, difference, symmetric difference, and their respective update variants—and demonstrates how they can be performed using **special Python operators** such as ``|``, ``&``, ``-``, and ``^``. Practical examples using Jupyter Notebook show how these operators work with sets, either producing new sets or updating existing ones in place. The lecture emphasizes that these operator forms provide a more concise and programmer-friendly syntax without changing the underlying semantics or outcomes of the operations.

Section 2: Key Concepts and Explanations

- Set Operations (Review)

- Union
- Intersection
- Difference
- Symmetric difference
- Update operations for these (modifying the original set)

- Operators Corresponding to Set Operations

Operation	Method	Operator	Update Operator (In-place)
Intersection	<code>set1.intersection(set2)</code>	<code>&</code>	<code>&=</code>
Difference	<code>set1.difference(set2)</code>	<code>-</code>	<code>-=</code>
Symmetric Difference	<code>set1.symmetric_difference(set2)</code>	<code>^</code>	<code>^=</code>

- How these operators work:

- **Union (`|`):** Combines all unique elements from both sets.
- **Intersection (`&`):** Gets only elements common to both sets.
- **Difference (`-`):** Gets elements in the first set not present in the second.
- **Symmetric difference (`^`):** Gets elements from both sets except those common to both.

- **Update operators (`|=`, `&=`, `-=`, `^=`)** perform the operation and store the result in the first set, modifying it in place.

- Benefits of Operators:

- More readable and concise syntax compared to method calls.
- Easy to remember and use interchangeably with methods.
- Helps write cleaner and more Pythonic code.

Section 3: Example Code and Use Cases

Creating two sets

Union operation - create a new set s3

Output: {1, 2, 3, 5, 7, 9, 10, 11}

Explanation: All unique elements from both s1 and s2 combined

Union update - modify s1 in place

Output: {1, 2, 3, 5, 7, 9, 10, 11}

Explanation: s1 updated to the union of s1 and s2

Reset sets for next example

Intersection operation - new set s3

Output: {5, 7}

Explanation: common elements in s1 and s2

Intersection update - modify s1 in place

Output: {5, 7}

Reset sets again

Difference operation - new set s3

Output: {1, 2, 3}

Explanation: elements in s1 not in s2

Difference update - modify s1 in place

Output: {1, 2, 3}

Reset sets

Symmetric difference operation - new set s3

Output: {1, 2, 3, 9, 10, 11}

Explanation: elements in either s1 or s2 but not both

Symmetric difference update - modify s1 in place

Output: {1, 2, 3, 9, 10, 11}

```
s1 = {1, 2, 3, 5, 7}
s2 = {5, 7, 9, 10, 11}
```

```
s3 = s1 | s2
print(s3)
```

```
s1 |= s2
print(s1)
```

```
s1 = {1, 2, 3, 5, 7}
s2 = {5, 7, 9, 10, 11}
```

```
s3 = s1 & s2
print(s3)
```

```
s1 &= s2
print(s1)
```

```
s1 = {1, 2, 3, 5, 7}
s2 = {5, 7, 9, 10, 11}
```

```
s3 = s1 - s2
print(s3)
```

```
s1 -= s2
print(s1)
```

```
s1 = {1, 2, 3, 5, 7}
s2 = {5, 7, 9, 10, 11}
```

```
s3 = s1 ^ s2
print(s3)
```

```
s1 ^= s2
print(s1)
```

These examples illustrate how the operators can be used to perform set operations and optionally update the original set.

Section 4: Key Takeaways

- Python sets support **operators** for union (`|`), intersection (`&`), difference (`-`), and symmetric difference (`^`).
- Each operator has an **update form** (e.g., `|=`) that modifies the set in place, similar to method variants like `.union_update()` etc.
- Using operators is a cleaner and often simpler alternative to calling set methods directly.
- The **results from operators and methods are identical**, so use whichever style fits your coding preference.
- Remembering operator symbols for set operations can speed up and simplify your code writing in Python.
- Practice these operations to confidently incorporate set operators in your projects and programs.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Operators on Set

Operators on Set

Union

Operators on Set

Union
Intersection

Operators on Set

Union
Intersection
Intersection update

Operators on Set

Union
Intersection
Intersection update
Difference

Operators on Set

Union
Intersection
Intersection update
Difference
Difference update

Operators on Set

Union
Intersection
Intersection update
Difference
Difference update
Symmetric difference

Operators on Set

Union
Intersection
Intersection update
Difference
Difference update
Symmetric difference
Symmetric difference update



Operators on Set

	Union
	Intersection
	Intersection update
	Difference
	Difference update
	Symmetric difference
	Symmetric difference update



Operators on Set

	Union
&	Intersection
	Intersection update
	Difference
	Difference update
	Symmetric difference
	Symmetric difference update

Operators on Set

	Union
&	Intersection
&=	Intersection update
	Difference
	Difference update
	Symmetric difference
	Symmetric difference update

Operators on Set

	Union
&	Intersection
&=	Intersection update
-	Difference
- =	Difference update
^	Symmetric difference
^=	Symmetric difference update

Operators on Set

&
&=
-
- =
^
^=

Operators on Set

|

&

&=

-

- =

^

^=

Operators on Set

localhost

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

{ }:

|

&

&=

-

- =

^

^=

Operators on Set

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[]:

S1

12357

S2

5791011

Operators on Set

localhost

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python 3 (ipykernel)

[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[]:

S1

12357

S2

5791011

Union

s3 = s1 | s2

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}

[2]: s3 = s1 | s2

[]: 
```

S112357

S2~~5~~~~7~~91011

S31235791011

Union

S3 = S1 | S2

|

&

&=

-

-=

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}

[2]: s3 = s1 | s2

[3]: s3

[3]: {1, 2, 3, 5, 7, 9, 10, 11}

[]: 
```

S112357

S2~~5~~~~7~~91011

S31235791011

Union

S3 = S1 | S2

|

&

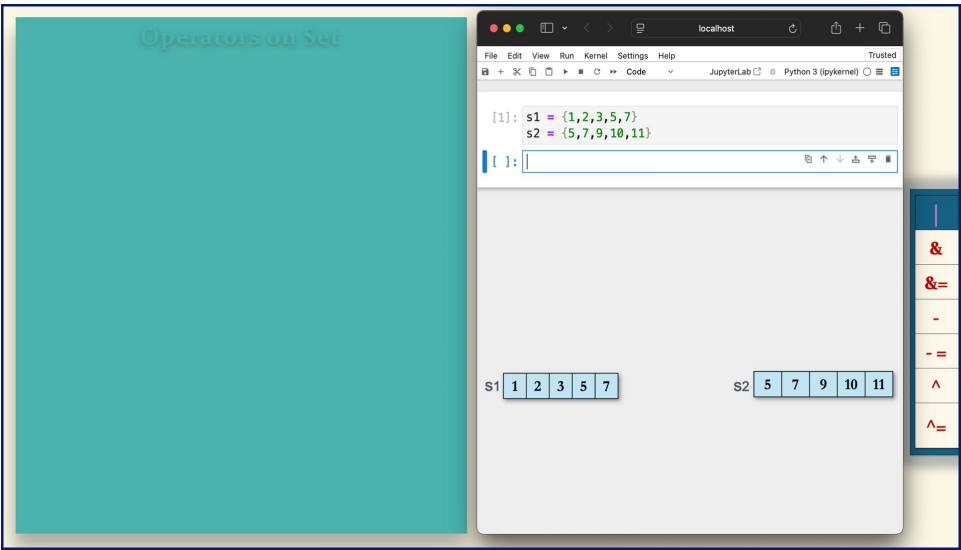
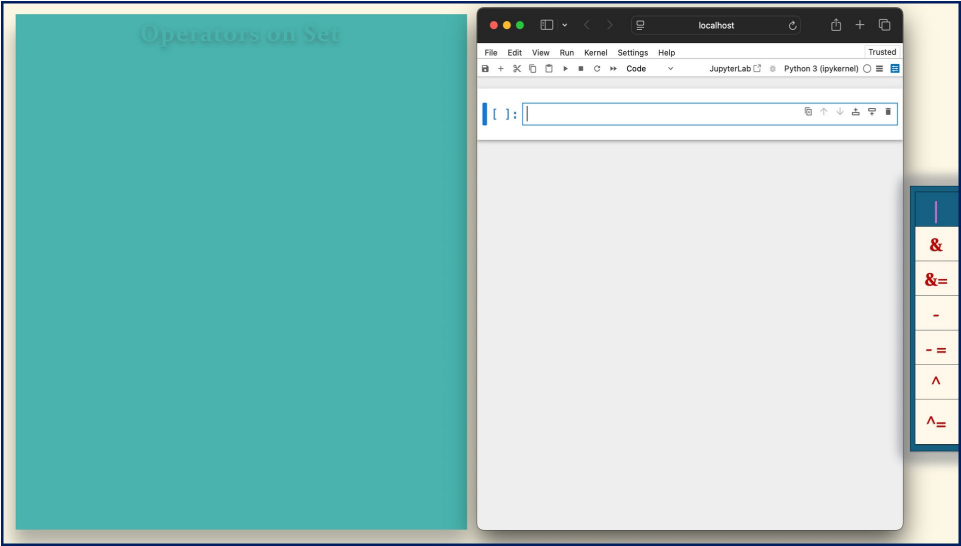
&=

-

-=

^

^=



Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}

[2]: s1 = s1 | s2

In [1]:
```

S1

1235791011

S2

5791011

Union

s1 = s1 | s2

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[ ]:
```

S112357

S25791011

Intersection

S3 = S1 & S2

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[2]: s3 = s1 & s2

[ ]:
```

S112357

S25791011

S357

Intersection

S3 = S1 & S2

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}

[2]: s3 = s1 & s2

[3]: print(s3)

[ ]:
```

S112357

S25791011

S357

Intersection

s3 = s1 & s2

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}

[ ]:
```

S112357

S25791011

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[]:
```

S1

12357

S2

5791011

Intersection

S1 &= S2

|

&

&=

-

-=

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[2]: s1 &= s2

[]:
```

S1

57

S2

5791011

Intersection

S1 &= S2

|

&

&=

-

-=

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}
```

S1

12357

S2

5791011

Difference

$S3 = S1 - S2$

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}
```

S1

12357

S2

5791011

Difference

$S3 = S1 - S2$

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}
[2]: s3 = s1 - s2
```

S112357

S25791011

S3123

Difference

$s3 = s1 - s2$

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}
```

S112357

S25791011

Difference

$s1 -= s2$

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}

[2]: s1 - s2
```

S1123

S25791011

Difference
S1 - S2

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
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[ ]:
```

S1

12357

S2

5791011

Symmetric Difference

$S3 = S1 \wedge S2$

Operators on Set

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[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[ ]:
```

S1

12357

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Symmetric Difference

$S3 = S1 \wedge S2$

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}

[2]: s3 = s1 ^ s2

In [ ]:
```

S112357

S25791011

S312391011

Symmetric Difference

$s3 = s1 \wedge s2$

|

&

&=

-

- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
      s2 = {5,7,9,10,11}

[2]: s3 = s1 ^ s2

[3]: print(s3)
{1, 2, 3, 9, 10, 11}

In [ ]:
```

S112357

S25791011

S312391011

Symmetric Difference

$s3 = s1 \wedge s2$

|

&

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- =

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Operators on Set

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[1]: s1 = {1,2,3,5,7}
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S1

12357

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Symmetric Difference

$S1 = S1 \wedge S2$

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&

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Operators on Set

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[1]: s1 = {1,2,3,5,7}
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[2]: s1 = s1 ^ s2
print(s1)
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S1

12391011

S2

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Symmetric Difference

$S1 = S1 \wedge S2$

|

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- =

^

^=

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}
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S1

12357

S2

5791011

Symmetric Difference

S1 ^ S2

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}
```

S1

12357

S2

5791011

Symmetric Difference

S1 ^ S2

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[2]: s1 ^ s2
```

```
{ 1: }
```

S1

12391011

S2

5791011

Symmetric Difference

S1 ^ S2

Operators on Set

```
[1]: s1 = {1,2,3,5,7}
s2 = {5,7,9,10,11}

[2]: s1 ^ s2

[3]: print(s1)

{1, 2, 3, 9, 10, 11}
```

```
{ 1: }
```

S1

12391011

S2

5791011

Symmetric Difference

S1 ^ S2

